

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

Mathematics — SSC-I (9th Grade)

Syllabus: Unit 4 — Factorization & Algebraic Manipulation | Unit 5 — Linear Equations & Inequalities
| Unit 7 — Coordinate Geometry

Total Marks: 65

Time Allowed: 2 Hours 30 Minutes

Version: 1

General Instructions:

- Section A** (MCQs): Attempt **ALL 12** questions. Each carries **1 mark**. Time: 15 minutes.
- Section B** (Short Questions): Attempt **11**, choosing main **OR** alternative. Each **3 marks**.
- Section C** (Long Questions): Attempt **4**, choosing main **OR** alternative. Each **5 marks**.
- Use black or blue ink only. Pencil is not allowed except for diagrams.

SECTION A — Multiple Choice Questions ($12 \times 1 = 12$ Marks)

Encircle the correct option. All questions are compulsory.

#	Question	A	B	C	D	Answer
1	The factorization of $x^4 + 4y^4$ is:	(x^2+2y^2+2xy) (x^2+2y^2-2xy)	(x^2+2y^2) (x^2-2y^2)	$(x^2+2y^2)^2$	$(x+y)^4$	☐ A ☒ B ☐ C ☐ D
2	Factors of $y^2 - 7y + 12$ are:	$(y - 3)(y - 4)$	$(y + 3)(y + 4)$	$(y - 12)(y + 1)$	$(y - 6)(y - 2)$	☐ A ☒ B ☐ C ☐ D
3	The HCF of $18a^3b^4c^5$ and $24a^2b^5c^3$ is:	$6a^2b^4c^3$	$6a^3b^5c^5$	$72a^3b^5c^5$	$6a^3b^4c^3$	☐ A ☒ B ☐ C ☐ D
4	The product of HCF and LCM of two polynomials equals:	Their sum	Their difference	Their product	Their quotient	☐ A ☒ B ☐ C ☐ D
5	The square root of $9a^2 + 12ab + 4b^2$ is:	$\pm(3a + 2b)$	$(3a + 2b)$	$\pm(3a - 2b)$	$3a \pm 2b$	☐ A ☒ B ☐ C ☐ D
6	The factorization of $a^3 + 27b^3$ is:	$(a + 3b)(a^2 - 3ab + 9b^2)$	$(a - 3b)(a^2 + 3ab + 9b^2)$	$(a + 3b)(a^2 + 3ab + 9b^2)$	$(a - 3b)(a^2 - 3ab + 9b^2)$	☐ A ☒ B ☐ C ☐ D
7	Which of the following is a linear equation in one variable?	$x^2 + 3 = 0$	$xy + 3 = 0$	$2x + 3 = 0$	$2x^2 + 3 = 0$	☐ A ☒ B ☐ C ☐ D
8	The solution set of $ x + 1 = 2$ is:	$\{1, -3\}$	$\{-1, 3\}$	$\{1, 3\}$	$\{-1, -3\}$	☐ A ☒ B ☐ C ☐ D
9	The solution set of $\sqrt{5x} = -10$ is:	$\{20\}$	$\{ \}$ (empty)	$\{-20\}$	$\{-2\}$	☐ A ☒ B ☐ C ☐ D
10	The distance between points A(-1, 3) and B(4, 15) is:	12	13	15	169	☐ A ☒ B ☐ C ☐ D
11	The minimum number of sides in a polygon is:	2	3	4	5	☐ A ☒ B ☐ C ☐ D
12	The midpoint of the line segment joining (2, 4) and (6, 10) is:	(3, 7)	(4, 6)	(4, 7)	(8, 14)	☐ A ☒ B ☐ C ☐ D

SECTION B — Short Questions ($11 \times 3 = 33$ Marks)

Attempt 11 questions. Choose either the main question OR its alternative.

Q.	Question	Marks	OR	OR Question	Marks
2(i)	Factorize: (a) $x^2 - 8x + 15$ (b) $3x^2 + 7x + 2$	3	OR	Factorize: (a) $x^2 - 4x - 5$ (b) $2x^2 + 9x + 4$	3
2(ii)	Factorize: $a^3 + 27$	3	OR	Factorize: $8x^3 - 125y^3$	3
2(iii)	Find the HCF of $12x^2y^3$ and $18xy^2$ by factorization.	3	OR	Find the HCF of $a^2 - 4$ and $a^2 + 4a + 4$ by factorization.	3
2(iv)	Find the LCM of $x^2 - 1$ and $x^2 + 2x + 1$.	3	OR	Find the LCM of $6a^2b$ and $8ab^3$.	3
2(v)	Find the square root of $4x^2 + 12xy + 9y^2$ by factorization.	3	OR	Simplify: $(x^2-4)/(x^2-5x+6) \times (x-3)/(x+2)$	3
2(vi)	Solve for x: $3x/5 - 1/2 = x/4 + 1$	3	OR	Solve for x: $5x - 2 - x = 4 - 3x - 27$	3
2(vii)	Solve the absolute value equation: $ 2x + 5 = 11$	3	OR	Solve the absolute value equation: $ 3x + 2 = 7$	3
2(viii)	Solve the inequality and write the solution set: $3(-4 + 5y) \leq -8(1 - 2y) + 6$	3	OR	Solve the inequality and write the solution set: $8x + 9 < 6x - 7$	3
2(ix)	Find the distance between the points $(-2, 1)$ and $(1, 5)$.	3	OR	Find the distance between the points $(0, -5)$ and $(-2, 1)$.	3
2(x)	Find the midpoint of the line segment joining $(2, 5)$ and $(6, 9)$.	3	OR	Find the midpoint of the line segment joining $(-1, 0)$ and $(2, 2)$.	3
2(xi)	Show that the points $A(3, 2)$, $B(4, 4)$, $C(5, 6)$ are collinear.	3	OR	If $(1, 4)$ is the midpoint of the segment joining $(6, b)$ and $(a, 2)$, find a and b .	3

SECTION C — Long Questions ($4 \times 5 = 20$ Marks)

Attempt 4 questions. Choose either the main question OR its alternative.

Q.	Question	Marks	OR	OR Question	Marks
3(i)	Factorize: $(x + 5)(x + 3)(x + 2)(x + 6) - 88$	5	OR	Factorize: $(a^2 - 5)^2 - 13(a^2 - 5) + 36$	5
3(ii)	Find the square root of $x^4 + 8x^3 + 20x^2 + 16x + 4$ by division method.	5	OR	Find the HCF of $12 + 16a + 7a^2 + a^3$ and $13a + 5a^2 + 14 + a^3$ by division method.	5
3(iii)	Solve the compound inequality: $2x + 3 > 7$ OR $4x - 1 < 3$ (where $x \in \mathbb{R}$). Write the solution set.	5	OR	Solve the compound inequality: $x - 5 \geq -1$ AND $x + 3 \leq 10$ (where $x \in \mathbb{R}$). Write the solution set.	5
3(iv)	Show that the points $A(-4, 2)$, $B(1, 4)$, $C(3, -1)$, $D(-2, -3)$ are the vertices of a square.	5	OR	Show that the points $A(-4, -1)$, $B(0, -2)$, $C(6, 1)$, $D(2, 2)$ are the vertices of a parallelogram.	5

Invigilator's
Signature

Candidate's
Signature

Chapters Covered: Unit 4 — Factorization & Algebraic Manipulation | Unit 5 — Linear Equations & Inequalities | Unit 7 — Coordinate Geometry

— End of Question Paper —